

Charge movement in a dipole field

Y13 (2013) — English translation

There is a rigidly fixed electric dipole (two opposite, but identical in module point charges $+q$ and $-q$, located at a distance $l = 2$ cm from each other). At a distance $L = 2$ cm from the dipole on a line perpendicular to its axis (Fig. 3), a ball of mass m , charged with a charge $+q$, is located at equal distances from the dipole charges. The ball is released without a push ($v_0 = 0$), and it begins to move in the electric field of the dipole.

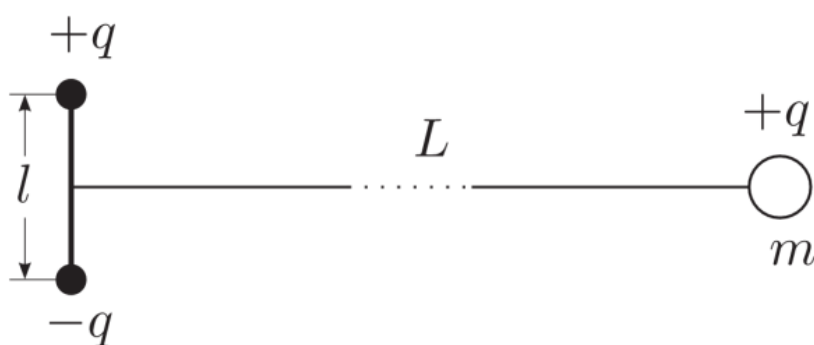


Figure 1: Figure 1.

A1	Show that the curve along which the ball will move is an arc of a circle.	0
A2	Indicate the points at which the ball will have maximum and minimum speeds.	0
A3	Find the maximum speed of the ball, taking the value $\frac{q^2}{4\pi\epsilon_0 m} = \frac{\text{V}}{\text{m} \cdot \text{kg}}$	0

Source: <https://pho.rs/p/3972>